Supplementary Material for “Coverage-Qualified Evaluation-Contract Auditing for Aerial Vehicle Detection”

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TABLE S1
DECLARED EVALUATION CONTRACTS. G IS ALL VALID GT, G_t THE SUPPORT-RETAINED TARGET, $D_t = G \setminus G_t$, AND H SOURCE IGNORE.

ID	Scored valid GT	Neutral regions	Step meaning
A	G	none	all-valid reference
F	G_t	none	removed objects are background
R	G_t	D_t	neutralize removed objects
S	G_t	$D_t \cup H$	add source ignores

TABLE S2
FIVE-CONFIGURATION VISDRONE PATH SUMMARY AT CONFIDENCE AND VALID-MATCH IOU .25. A = ABSOLUTE 24 PX; N = NORMALIZED .015.

Configuration	Rule	δ_{target}	δ_{removed}	δ_{source}	Range
B-640	A	$-.0360$	$+.0667$	$+.0093$	.0760
	N	$+.0178$	$+.0093$	$+.0078$	.0349
B-1280	A	$-.0930$	$+.0911$	$+.0281$	.1192
	N	$-.0192$	$+.0252$	$+.0237$	.0489
Y11m	A	$-.0310$	$+.0727$	$+.0119$	.0846
	N	$+.0254$	$+.0088$	$+.0100$	.0441
ASD	A	$-.0759$	$+.0932$	$+.0166$	.1099
	N	$-.0025$	$+.0203$	$+.0138$	.0341
Y11n-P2	A	$-.0699$	$+.0922$	$+.0159$	.1080
	N	$-.0032$	$+.0216$	$+.0131$	.0348

S1. DECLARED CONTRACTS AND COUNT LEDGER

The four contracts alter one evaluator factor at a time. All detections are score ordered; retained valid GT is matched before an otherwise unmatched detection can be neutralized. Removed and source-ignore regions use intersection over prediction area at .5. Table S1 distinguishes the scored target from neutral regions explicitly.

Table S2 summarizes the five support-conditioned count-transition ledgers behind Fig. 2 of the main paper. Every row comes from one frozen prediction artifact. The released 90-row ledger contains the corresponding integer TP/FP/FN changes. Each three-step sum equals $F_{1,S} - F_{1,A}$ to machine precision.

Table S3 extends the path beyond one baseline per source. VisDrone contributes five separately materialized artifacts. UAVDT contributes three resolution slices and a tiling configuration. AI-TOD contributes three resolution slices on the same covered subset and estimates cache-resolution sensitivity for that 1,869-image population.

Removed-object neutralization is positive in all 24 configuration–rule replays. Source-ignore handling is positive in all ten VisDrone replays and exactly zero for UAVDT and AI-TOD, which have no active source-ignore regions. Target filtering remains configuration dependent, especially under normalized support. The exact 96-row endpoint record and 72-row transition ledger preserve artifact groups and input hashes.

S2. SCENE-STRATIFIED PATHS AND TASK CONSEQUENCES

The strata use annotation-derived tertiles fixed before metric aggregation. Density includes all 548 images. Target-size and occlusion strata include 519 nonempty images; nearest-neighbor strata include 510 images with at least two vehicles. The density cutpoints are 17 and 37 vehicles/image, target-size cutpoints are 35 and 55 pixels, nearest-neighbor cutpoints are .0326 and .0565, and occlusion-fraction cutpoints are .400 and .593.

Small-target, crowded, and dense scenes concentrate the absolute support response. Normalization attenuates each contrast while preserving the same ordering for target size, neighbor spacing, and density. Occlusion follows a nonmonotonic path gradient. The Y11m–ASD point difference remains positive in every one of the 24 complete factor–tertile–rule cells.

The operational record separates false-alarm burden from missed-target burden. Y11m improves both components under 24-pixel support. Under normalized support it produces fewer false alarms and higher review yield, while ASD produces fewer misses per 100 scored vehicles.

S3. COVERAGE QUALIFICATION AND FULL SUPPORT GRID

Raw prediction names establish coverage before class filtering. VisDrone and UAVDT cover all declared images. AI-TOD covers 1,869/2,804 images. Its full-annotation support diagnosis uses all 2,804 images, and the metric replay targets the 1,869-image covered subset. The covered images contain 16,900 boxes; the 935 noncovered images contain 43,004.

The image-level record establishes distinct covered-subset and full-validation distributions. The covered subset has many empty vehicle images, and its nonempty images also have a different support composition. At the box level, covered and noncovered median max sides are 12 and 16 pixels; the normalized-support Wasserstein distance is .00370. Metric endpoints target the observed covered-set population.

S4. PAIRED RANK RECORDS AND EVALUATOR CONTROLS

Table S8 gives the localization-by-metric control for the Y11m–ASD pair. Sequence clusters are the leading seven-digit VisDrone filename tokens (76 clusters). Every replicate includes all images in each sampled sequence, applies identical multiplicities to both configurations, and recomputes global

TABLE S3
CROSS-CONFIGURATION CONTRACT-PATH RANGES AT CONFIDENCE AND VALID-MATCH IOU .25. EACH INTERVAL SPANS THE CONFIGURATIONS WITHIN ONE SOURCE. A = 24 PX; N = .015. AI-TOD IS CONDITIONED ON 1,869/2,804 IMAGES.

Source	Rule	Configs.	δ_{target}	$\delta_{\mathrm{removed}}$	δ_{source}	Contract range
VisDrone	A	5	$-.0930$–$-.0310$	$+.0667$–$+.0932$	$+.0093$–$+.0281$	.0760–.1192
	N	5	$-.0192$–$+.0254$	$+.0088$–$+.0252$	$+.0078$–$+.0237$	.0341–.0489
UAVDT	A	4	$-.0147$–$+.0001$	$+.0029$–$+.0067$	0	.0030–.0147
	N	4	$-.0041$–$+.0025$	$+.0010$–$+.0027$	0	.0018–.0041
AI-TOD[a]	A	3	$-.5105$–$-.4565$	$+.1450$–$+.1920$	0	.4565–.5105
	N	3	$-.1620$–$-.0943$	$+.0859$–$+.1028$	0	.0943–.1620

[a] Coverage-conditioned resolution control; all three slices share one checkpoint and covered image set.

TABLE S4
ENDPOINT SCENE STRATA FOR VISDRONE. PATH QUANTITIES ARE MEDIANS ACROSS FIVE CONFIGURATIONS; PAIR IS TERMINAL-S FIXED-CONFIDENCE F_1(Y11M)$-F_1$(ASD). MIDDLE-TERTILE RECORDS ARE RELEASED. A = 24 PX; N = .015.

Rule	Factor	Stratum	δ_{target}	$\delta_{\mathrm{removed}}$	Contract range	Pair
A	Density	Sparse	$-.0507$	$+.0532$	.0721	$+.0236$
		Dense	$-.0818$	$+.1107$	.1252	$+.0250$
	Target size	Small	$-.1558$	$+.1771$	.2121	$+.0242$
		Large	$-.0047$	$+.0056$	.0131	$+.0152$
	Neighbor spacing	Crowded	$-.1188$	$+.1459$	.1682	$+.0297$
		Dispersed	$-.0164$	$+.0186$	.0346	$+.0125$
	Occlusion	Low	$-.0670$	$+.0820$	.0955	$+.0155$
		High	$-.0687$	$+.0912$	.1042	$+.0287$
N	Density	Sparse	$-.0021$	$+.0089$	.0273	$+.0220$
		Dense	$-.0026$	$+.0269$	.0406	$+.0154$
	Target size	Small	$-.0086$	$+.0438$	.0617	$+.0102$
		Large	$-.0006$	$+.0006$	.0074	$+.0151$
	Neighbor spacing	Crowded	$-.0067$	$+.0378$	.0540	$+.0144$
		Dispersed	$-.0002$	$+.0015$	.0166	$+.0127$
	Occlusion	Low	$-.0054$	$+.0185$	.0325	$+.0143$
		High	$-.0006$	$+.0185$	.0334	$+.0177$

TABLE S5
TERMINAL-S TASK CONSEQUENCES AT CONFIDENCE AND VALID-MATCH IOU .25. YIELD IS CONFIRMED VEHICLES PER 100 SCORED ALERTS.

Rule	Candidate	FP/image	FN/100 vehicles	Yield
A	Y11m	3.01	8.76	87.38
	ASD	3.93	9.80	83.96
N	Y11m	3.03	12.36	88.99
	ASD	4.32	11.57	85.10

TABLE S6
AI-TOD COVERED–NONCOVERED IMAGE QUALIFICATION. BRACKETS ARE MEDIAN [Q1,Q3]. SMD USES COVERED MINUS NONCOVERED MEANS; W_1 IS IN THE FEATURE'S NATIVE UNIT.

Feature	Covered	Noncovered	SMD	W_1
Boxes/image	0 [0,1]	21 [7,58.5]	$-.673$	36.951
Median side (px)[a]	17 [12,22]	16 [14,18]	$+.448$	5.597
Fraction < 24 px[a]	.895 [.600,1]	.961 [.857,1]	$-.640$	.159
Fraction $< .015$[a]	.018 [0,.410]	.009 [0,.111]	$+.610$	.148
Nearest distance[b]	.031 [.016,.065]	.027 [.018,.055]	$+.029$	.009

[a] Nonempty images. [b] Images with at least two vehicle boxes; centers normalized by image size.

TABLE S7
COMPLETE POOLED RETAINED VALID VEHICLE GT (%) BEFORE PREDICTION MATCHING.

Rule	Threshold	VisDrone	UAVDT	AI-TOD
Absolute	16 px	85.7	99.6	47.6
	20 px	79.5	98.8	20.4
	24 px	73.3	95.1	9.7
	28 px	67.5	77.5	5.8
	32 px	62.1	56.4	4.2
	40 px	52.3	34.3	2.5
	48 px	42.7	30.0	1.4
	64 px	27.7	17.7	0.4
Normalized	.0050	100.0	100.0	99.9
	.0075	98.8	100.0	99.6
	.0100	96.5	99.8	98.2
	.0150	89.8	97.4	81.0
	.0200	82.4	80.0	47.6
	.0300	67.6	46.4	9.7

TABLE S8
Y11M–ASD DIFFERENCES UNDER TERMINAL CONTRACT S (10,000 PAIRED SEQUENCE-CLUSTER REPLICATES). A = 24 PX; N = .015.

Rule	IoU/metric	Diff.	Pointwise 95%	Simultaneous 95%	p_{adj}
A	.25/max-F_1	$+.0163$	[.0119,.0213]	[.0097,.0230]	$< .001$
	.25/AP	$+.0081$	[.0051,.0120]	[.0033,.0129]	$< .001$
	.50/max-F_1	$+.0141$	[.0099,.0191]	[.0076,.0207]	$< .001$
	.50/AP	$+.0013$	[$-.0033$,.0052]	[$-.0046$,.0072]	.984
N	.25/max-F_1	$+.0099$	[.0060,.0140]	[.0043,.0155]	$< .001$
	.25/AP	$-.0005$	[$-.0030$,.0047]	[$-.0061$,.0051]	1.000
	.50/max-F_1	$+.0054$	[.0013,.0103]	[$-.0010$,.0117]	.125
	.50/AP	$-.0110$	[$-.0181$,$-.0057$]	[$-.0199$,$-.0022$]	.008

counts or pooled-detection AP. The max-t critical value is 2.744 across the eight primary comparisons.

Fixed-confidence F_1 has positive pointwise intervals in all four cells and remains an operational secondary endpoint. The simultaneous family resolves three max-F_1 cells and two AP cells. Normalized max-F_1 at IoU .50 joins the two previously unresolved AP cells after multiplicity adjustment.

The alternative definitions preserve every max-F_1 and AP direction in the 2×2 design. Both neutralization steps remain positive in all 30 candidate–rule combinations.

The AP cap is 2,000 detections per image. The observed maximum is 1,881, placing every pool image below the cap; caps of 100 and 300 alter dense-image inputs. The independent AP implementation and COCOeval agree exactly for AP50 (maximum absolute difference $< 10^{-12}$). AP25 keeps the S-contract neutralization threshold fixed at .5, whereas COCOeval couples crowd matching to the evaluated IoU; the released controlled implementation supplies the AP25 endpoint. A threshold-first Hungarian replay selects the same top configuration as greedy matching in 24/24 settings; the maximum absolute F_1 difference is .00546.

The full 14-support-by-eight-confidence rank surface maps

TABLE S9
RETENTION-MATCHED SUPPORT-DEFINITION CONTROLS. RANGE IS THE SPAN OF FIVE CANDIDATE-LEVEL $A \to F \to R \to S$ RANGES. REMAINING COLUMNS ARE Y11M–ASD TERMINAL-S POINT DIFFERENCES.

Coordinate	Support definition (threshold)	Retained	Path range	max-$F_{1,25}$	AP25	max-$F_{1,50}$	AP50
Absolute	max side (24 px)	73.33%	.076–.119	$+.0163$	$+.0081$	$+.0141$	$+.0013$
	minimum side (17 px)	73.95%	.067–.115	$+.0174$	$+.0072$	$+.0152$	$+.0019$
	geometric mean (20.149 px)	73.30%	.073–.119	$+.0178$	$+.0083$	$+.0161$	$+.0025$
Normalized	max side (.015)	89.77%	.034–.049	$+.0099$	$-.0005$	$+.0054$	$-.0110$
	minimum side (.009559)	90.32%	.032–.047	$+.0088$	$-.0025$	$+.0045$	$-.0157$
	geometric mean (.012187)	89.77%	.034–.049	$+.0103$	$-.0009$	$+.0058$	$-.0113$

TABLE S10
MINIMAL REPLAY CHAIN IN THE RELEASED REPOSITORY.

Audit object	Released record or script
Coverage gate	`outputs/coverage_qualification/` and `evaluate_aitod_coverage_qualification.py`
Support denominator	`outputs/scale/box_scale_records.parquet` and raw-coverage manifests
Contract path	`outputs/contract_path/`, `contract_path.py`, and `evaluate_contract_path_headlines.py`
Metric-qualified rank	`outputs/metric_qualified_rank/` and `evaluate_metric_qualified_visdrone_rank.py`
Metric-factorial control	`outputs/metric_factorial_control/` and `evaluate_metric_factorial_control.py`
Figures and tests	Four `build_*.py` scripts; contract, support, matching, and AP regression tests
Repository	https://github.com/Marchematics/aerial-evaluation-audit

the sensitivity boundary. Confirmatory inference uses the eight prespecified named-pair cells and their simultaneous max-t intervals. Machine-readable full surfaces, matching controls, maxDets diagnostics, and DOTA-v2 structural support distances accompany the release.

S5. COVERAGE AND CAP CONTROLS

Figure 1 separates two boundary checks that precede metric interpretation. The AI-TOD density gap is visible at the image level, while the box-level curves show that coverage selection also changes the support composition of the observed target. The cap panel uses mapped vehicle predictions before AP truncation: a low cap would modify the prediction set for every configuration by a different amount. The declared cap of 2,000 exceeds the observed per-image maximum of 1,881 and therefore preserves every mapped prediction.

Figure 2 provides the structural context summarized outside the main figures. Normalization changes the coordinate system, compresses the numerical scale, and retains four distinct empirical distributions. The largest normalized pairwise distance is .031 between VisDrone and AI-TOD, whereas VisDrone and DOTA-v2 differ by .006. These values quantify distributional support differences.

S6. REPRODUCIBILITY RECORD

The data-free release provides configurations, derived records, scripts, and SHA-256 checksums. Licensed imagery, trained weights, and raw prediction caches remain external and are referenced by manifest path and hash. Reproduction order is coverage $\to$ support $\to$ contract endpoints/ledger $\to$ paired metric qualification $\to$ figures.

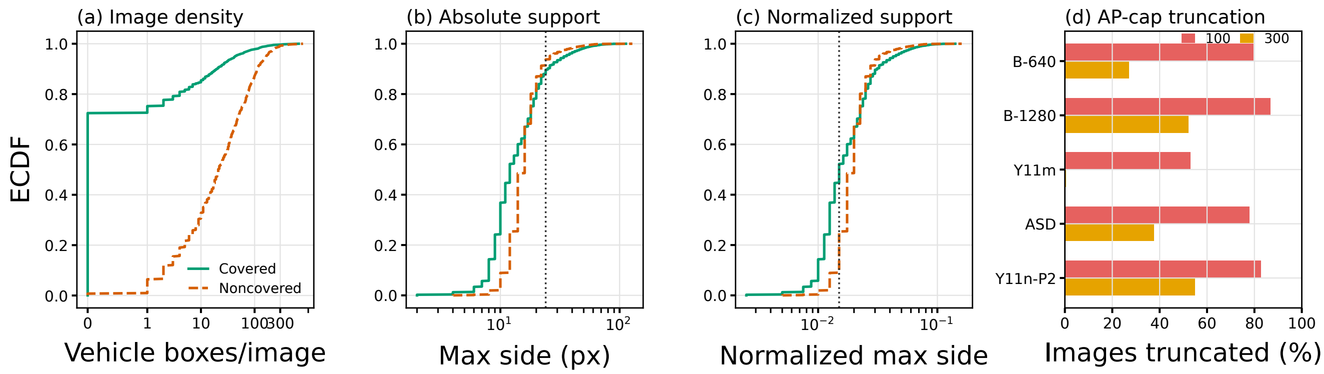

Fig. 1. Coverage and cap qualification records. (a)–(c) AI-TOD covered and noncovered image groups differ in density and support; vertical lines mark 24 px and .015. ECDFs in (b) and (c) use individual vehicle boxes. (d) Fractions of the 548 common VisDrone images whose mapped prediction counts exceed maxDets 100 or 300. At maxDets 2000, all five fractions are zero.

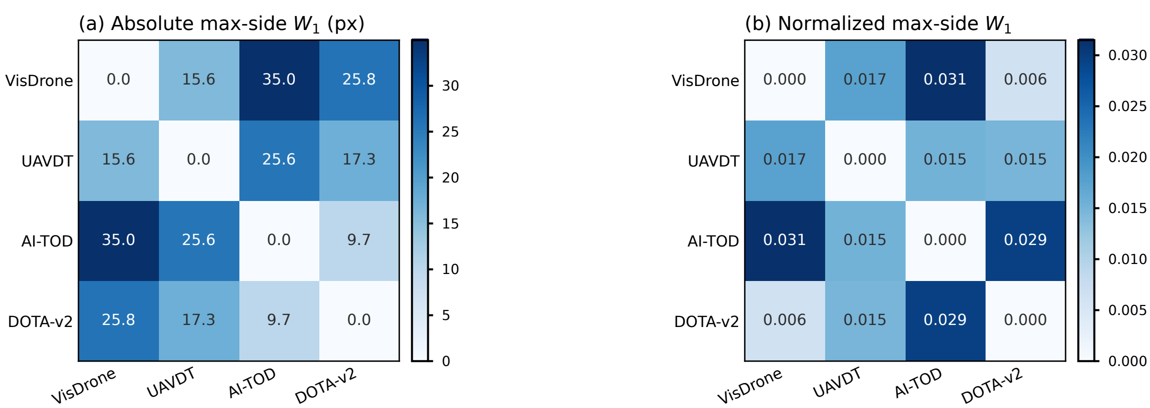

Fig. 2. Pairwise empirical Wasserstein distances for full-annotation vehicle support. (a) Absolute max side is scale covariant. (b) Normalized max side is resize invariant and retains nonzero source distances. DOTA-v2 supplies structural evidence; prediction-conditioned metrics and ranks use sources with available prediction artifacts.

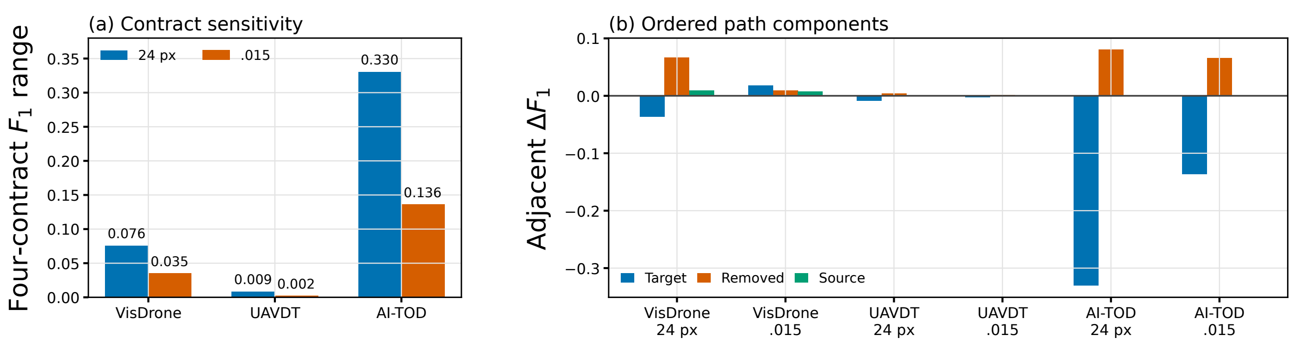

Fig. 3. Cross-source summary of the declared headline path. (a) Four-contract micro-F_1 ranges. (b) Adjacent path changes along the declared evaluation order. Normalized .015 reduces the range in all three sources and retains source-dependent residual sensitivity.